



DEALER BEST PRACTICE SERIES

793 Transmission Remove
& Install Tool

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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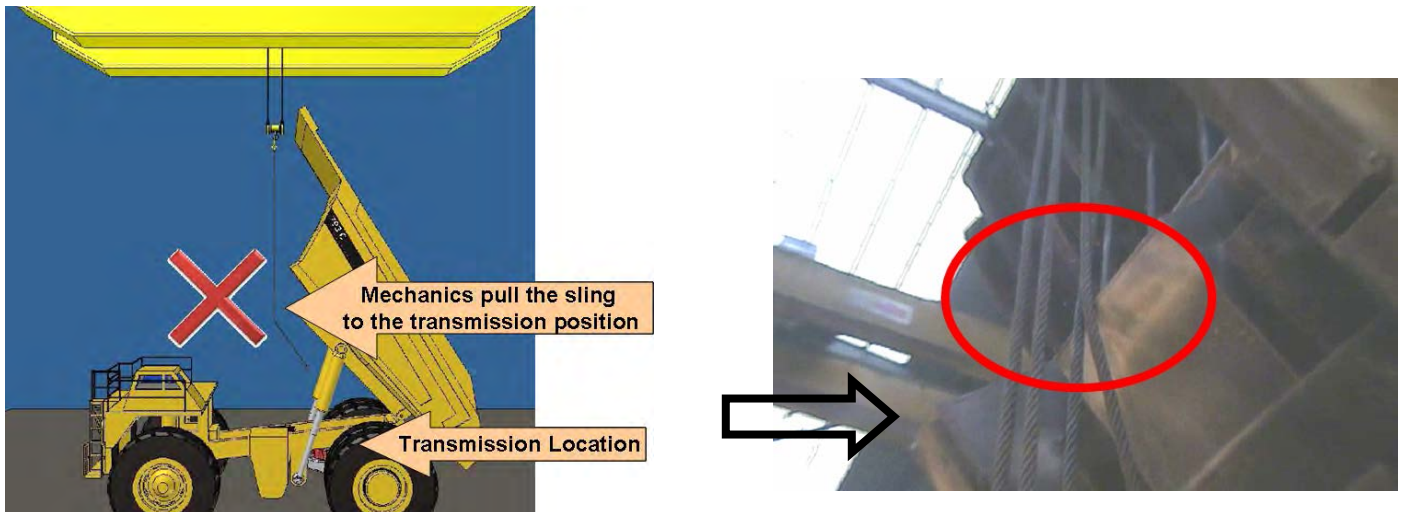
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1.0 Introduction

The ideal method to removal or install a transmission in an Off Highway Truck is to remove the truck body. This method provides clear access for an overhead crane to lift the transmission during the removal and installation process.

It is not always practical to remove the truck body, however, with the body on the truck, the body interferes with the lifting cable and it is not possible to lift and position the transmission properly for installation.



Truck Body Interferes with the Hoist Cables

In addition, transmissions should only be lifted using the lifting eye locations on the center of the transmission housing. Using other locations such as the rear of the housing can cause cracks in the housing.



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2/18/2015

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0812-3.04-1228

Trakindo has developed a fixture comprised of a rail and a lifting cage to perform removal installation without removing the body. The rail is positioned on the truck frame. The cage, equipped with three hand hoists that hold the transmission, is placed on the rail. The cage then can be moved to the proper position and the transmission is raised or lowered using the cage hoists.

This Best Practice is a Quick Win project.

2.0 Best Practice Description

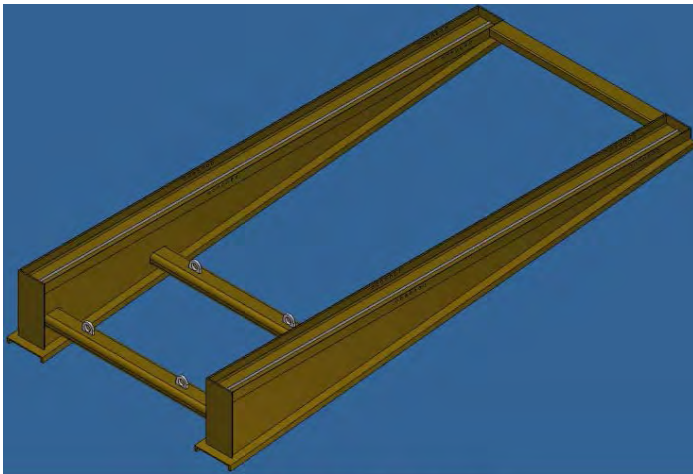
This fixture is a lifting device and needs to be designed and built to conform to local regulations and requirements.

Dimensions will depend on truck model, and may not work with “F” series trucks.

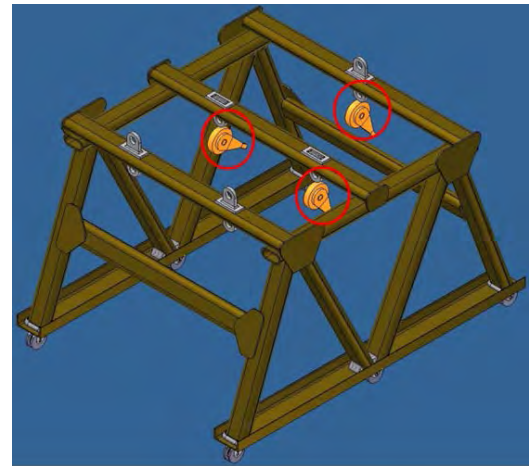
Tool Concept

Rail: To be placed on the truck frame.

Cage: Equipped with three one ton or higher hoists. The cage is lowered on to the rail with an overhead crane, and then moved along the rail using a ratchet puller to the transmission compartment location. Hoists are used to raise or lower the transmission.



THE RAIL



THE CAGE

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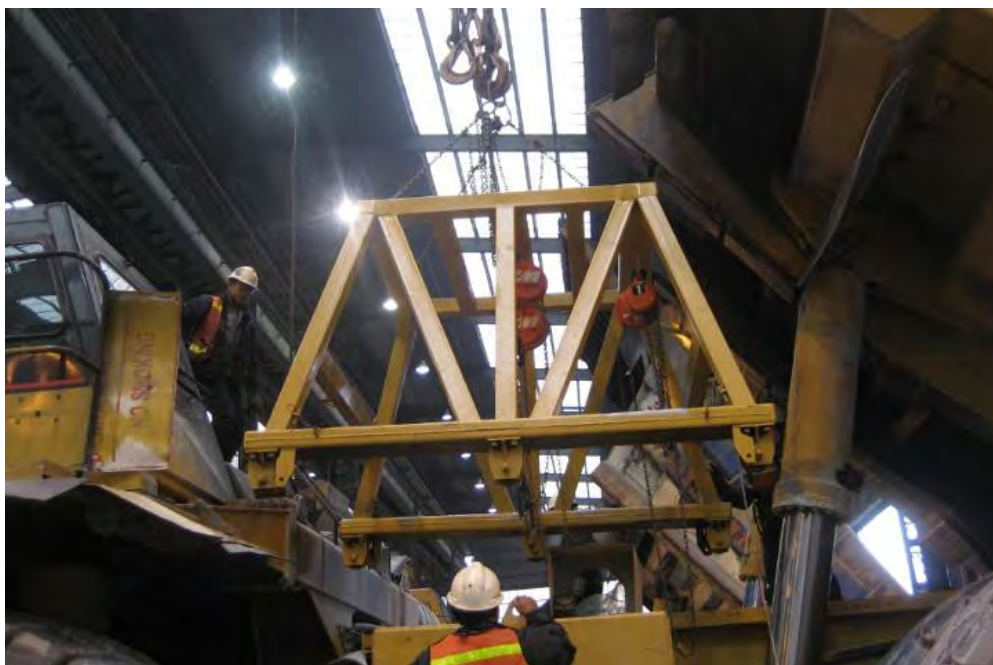
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Process steps

1. Installing the rail on the frame



2. Installing the cage on the rail



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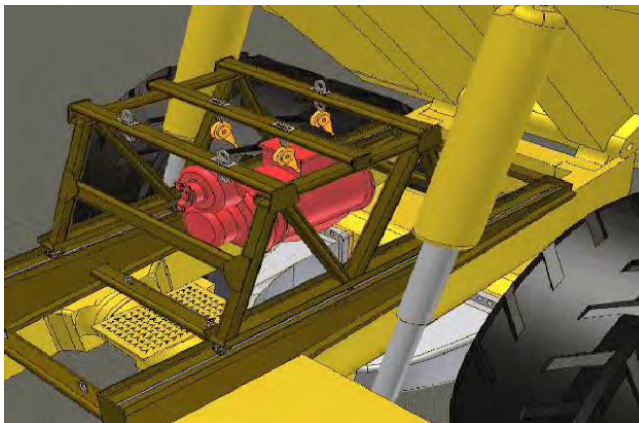
3. Positioning the cage above the transmission



4. Lifting up the transmission by using hand hoist



5. Moving the transmission group to the Overhead Crane lifting position using a ratchet puller.



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6. Lifting up the transmission group by using Overhead Crane



7. Placing the transmission group on shipping / storage stand.



3.0 Implementation Steps

- Design and fabricate the fixture. See design prints attached.
- Document removal and install process. See process attached.
- Train technicians.

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4.0 Benefits

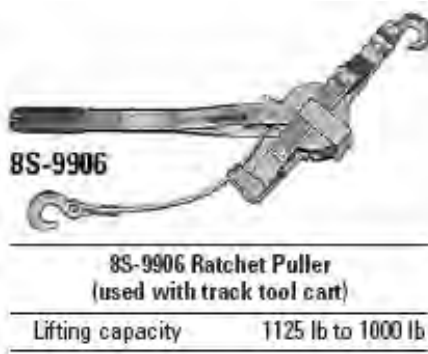
- Improved lifting and handling process.
- Reduced time required to remove and install a transmission

Activity	Time Needed (minutes)
Installing Rail	10 minutes
Installing Cage	6 minutes
Positioning Cage	5 minutes
Removing Transmission	14 minutes
Positioning Cage	5 minutes
Lift the Group	10 minutes

TOTAL TIME NEEDED 50 minutes

5.0 Resources Required

- o Costs to design and fabricate the fixture (diagram attached)
- o 25 tons (or more) over head crane
- o Three hand hoists (at least 1 ton each) to lift the transmission. (1U-9207)
- o 8S-9906 ratchet puller to pull the transmission group



6.0 Supporting Attachments / References

- o All videos available upon request to contact below
 - o Previous activity when 793 transmission removing process video.
 - o 793 Transmission R&I Tool Concept video.
 - o First time trial using the tool to remove 793 transmission video.

7.0 Related Best Practices

8.0 Acknowledgements

This Best Practice was written by:

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PRODUCT SUPPORT STANDARDS & DEVELOPMENT

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